Group Project 6: Auto Research for Materials Discovery

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1. Background

Autonomous materials discovery can be formulated as a closed-loop sequential decision problem. An AI system proposes candidate materials, receives feedback from a scientific oracle, and uses the accumulated evidence to decide what to investigate next. The challenge is not only to predict material properties, but to allocate a limited research budget toward the most informative and promising experiments.

MADE (MAterials Discovery Environments) is an open-source benchmark for end-to-end autonomous materials discovery. It provides closed-loop environments, scientific oracles, modular agent components, baseline systems, and discovery metrics. This project uses MADE as a fixed scientific environment and asks students to improve the autonomous research strategy rather than the underlying materials-science models.

This is an AI systems project, not a materials-science modeling project. Students are not expected to have prior materials-science knowledge; the benchmark environment and oracle provide the scientific feedback required by the agent.

2. Problem Description

Design and implement an Auto Research agent that maximizes materials-discovery performance on MADE under a fixed oracle-query budget. The benchmark environment, scientific oracle, base model and inference settings, validity checks, and final evaluator are held fixed. The optimization surface is the research system itself:

- **Planning and decomposition:** how the agent turns the discovery objective into sequential research actions.
- **Memory and evidence management:** how previous candidates, oracle feedback, failures, and discoveries are summarized and reused.
- **Exploration versus exploitation:** how the agent balances searching unfamiliar regions against refining promising ones.
- **Candidate proposal and tool routing:** how the agent selects among available generators, filters, scorers, and scientific tools.
- **Reflection and failure recovery:** how the agent detects unproductive search patterns and changes strategy.
- **Self-improvement or multi-agent research:** optional mechanisms that revise the research policy, prompts, memory, or division of labor during a run.

The benchmark oracle and evaluator must not be modified, and hidden ground-truth values must never be read before a candidate is queried. Fine-tuning the oracle, hard-coding held-out systems, or using benchmark labels outside the official interaction loop is not allowed.

The central question is whether a better autonomous research strategy can discover more scientifically valid materials from the same tools, base model, and research budget.

3. Evaluation

3.1 Metrics

- **Main metric - mSUN@B:** the benchmark mSUN score achieved under a fixed oracle-query budget B. Higher is better.
- **AUDC:** Area Under the Discovery Curve, measuring how quickly useful discoveries accumulate as the query budget is consumed.
- **AF and EF:** the benchmark Acceleration Factor and Enhancement Factor metrics, reported using the official MADE implementation.

- **Research cost:** total oracle calls, LLM/API tokens or dollars, and wall-clock time used by the Auto Research system.

3.2 Benchmark and Protocol

- **Benchmark:** MADE (MAterials Discovery Environments), using the official benchmark runner and unmodified scientific oracle/evaluation code.
- **Official repository:** https://github.com/diffractivelabs/MADE
- **Paper:** https://arxiv.org/abs/2601.20996
- **Benchmark runner:** use the official scripts/run_benchmark.py workflow. The course will freeze a repository commit, environment configuration, oracle, and set of materials systems before final evaluation.
- **Development/test split:** students may optimize on a public development set. Final ranking should use held-out systems selected by the teaching team from the same open benchmark environment.
- **Fair comparison:** all methods use the same base model, oracle, query budget, tool set, benchmark systems, and evaluator. Three repeated runs per method are preferred; one run is acceptable if compute is limited and this is stated.

4. Deliverables

Please submit the following deliverables via a Github repo or a Google Drive folder.

4.1 Codebase

- **An Auto Research agent:** integrated with the official MADE benchmark interface and capable of completing full closed-loop discovery episodes.
- **At least one developed research strategy** of your own design over the optimization surface in Section 2.
- **Required baselines for comparison:**
  - **Random-generator baseline:** the official MADE random generator baseline.
  - **Chemeleon generative baseline:** the official MADE Chemeleon baseline.
- **Reproduction material:** frozen configs, run commands, random seeds, dependency versions, and raw trajectories/results behind every reported number.

4.2 Report

The report must contain, at minimum:

- **Leaderboard results:** mSUN@B, AUDC, AF, EF, and research cost for the developed method and both required baselines.
- **Discovery curves:** cumulative discovery performance versus oracle-query budget.
- **Component ablation:** isolate the effect of planning, memory, reflection, exploration policy, multi-agent coordination, or other major components.
- **Generalization analysis:** compare performance on development systems and held-out test systems.
- **Failure classification:** categorize at least two failure modes with one trajectory example for each:
  - **Over-exploitation:** the system repeatedly searches near known successes and misses unexplored regions.
  - **Wasteful exploration:** the system spends too much budget on diverse but unproductive candidates.
- **Discussion:** explain which Auto Research mechanisms contributed most to discovery efficiency and where the remaining bottlenecks come from.

4.3 Presentation

Each group will give a **10-minute presentation**, followed by **5 minutes of Q&A**. The presentation should cover:

- The main idea and architecture of your Auto Research system.
- How the system interacts with MADE and manages the research budget.
- The most important benchmark results against the required baselines.
- Representative research trajectories, discoveries, and failure cases.
- The main conclusions and lessons learned from autonomous scientific discovery.